SECP 2613

SYSTEM ANALYSIS AND DESIGN

TITLE :

BASICS OF REQUIREMENTS ANALYSIS AND DESIGN TECHNIQUES

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ABSTRACT

Ensuring consumer satisfaction and business success requires designing systems and products that fulfill user needs. However, there is a gap in the literature regarding practical application of requirements analysis and design techniques to achieve this goal. By examining the fundamentals of requirements analysis and design techniques and their importance in user-centered design, this study aims to fill this knowledge gap.

The objectives of this research are to present the foundation of requirements analysis and design techniques, emphasize their significance in the design process, and exhibit their effectiveness in fulfilling user requirements. The main focus is on how requirements analysis and design techniques may be used to determine user needs, and support user-centric design processes.

In conclusion, this study emphasizes the significance of requirements analysis and design techniques in designing for user satisfaction. The research findings have significance for designers, developers and organizations who want to improve product usability and emphasize user happiness. Further studies have focused on analyzing the long-term impact of user-centered design techniques on product development and user engagement.

INTRODUCTION

Techniques are vital tools that engineers and practitioners use to negotiate complexity, reduce risks, and enhance solutions in the complex ecosystem of system design and analysis.

This paper offers a comprehensive study of system design and analysis techniques, explaining their importance, suitability, and influence on the development process as a whole. By using a varied range of known methods, the discussion seeks to provide readers with a comprehensive grasp of how techniques in system design and analysis are applied to overcome challenges and inspire innovation across many industries.

Furthermore, this paper aims to show the techniques in fixing real-world difficulties faced by system designers and analysts. Readers will be equipped with the knowledge and skills necessary to carefully negotiate the difficult landscape of system design

Fundamentally, system design and analysis techniques form the foundation for current engineering projects, acting as agents for sustainability, efficiency, and creativity. In the discussion that follows, this paper aims to shed light on the process of becoming proficient in system design and analysis techniques, which can lead to positive change and upgrade engineering practice.

1. Requirement analysis

   Requirement analysis techniques are techniques used in system design and analysis to extract human information requirements from members of the organization. There are three types of techniques: interviews, joint application design (JAD), and questionnaire-based surveys. There is a specific process for you to follow when communicating with users when using any of the three interactive information collection techniques. The specifics for every technique are listed below:

   1. Interviewing

      A focused, question-and-answer exchange with a specified goal is known as a data collection interview. There are 5 steps in interview preparation;

      1. Go over the background information.
      2. Set up the interviewing tools.
      3. Select the interviewee.
      4. Get the interviewee ready.
      5. Choose the question formats and types.

Writing the interview report

The work on the interview data doesn't end when the interview is over. To make the interviewer's perspective more understandable, the data must be written into a report. At a follow-up meeting, go over the interview report with the respondent. This stage assists in highlighting the meaning the interviewee intended to convey

2. JAD

IBM created JAD as an alternative to doing one-on-one user interviews. The goal of implementing JAD is to reduce the amount of time needed for personal interviews, enhance the quality of information requirements assessment outcomes, and increase user identification with newly developed information systems as a consequence of participatory procedures. With JAD, the system analyst can complete requirements analysis and collaborate with users in a group environment to develop the user interface.

When to use JAD

- User groups are impatient and seek creative solutions to common problems rather than traditional ones.
- The culture of the company encourages cooperative problem-solving among staff members at all levels.

- The company workflow allows for the absence of critical staff members for a period of two to four days.

3. Using questionnaire

System analysts can examine the views, beliefs, behaviours, and features of important individuals within the organisation who could be impacted by the present and planned systems by using questionnaires as a method of gathering information.

Writing questions

-open ended question

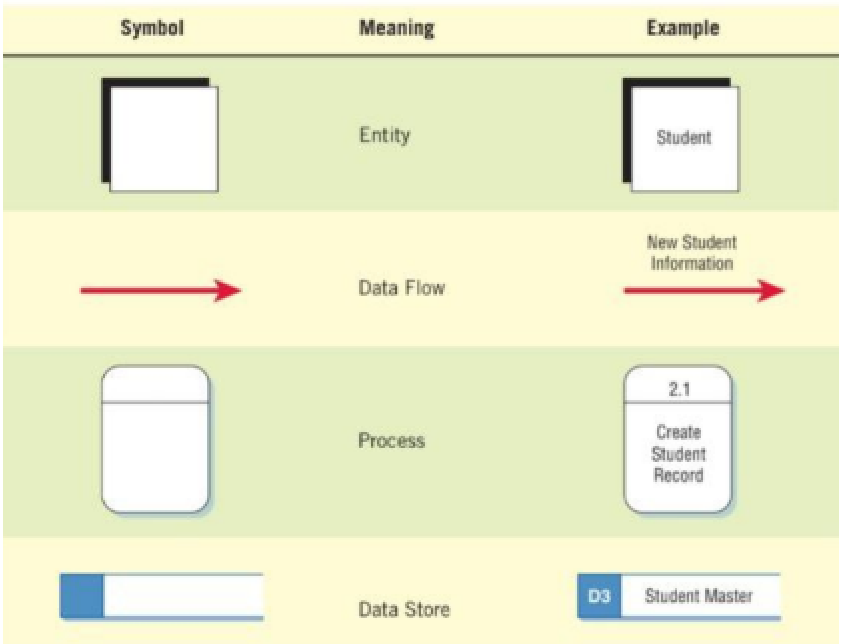
Open-ended questions are used when the system analyst want to know what people of the organisation think about a certain system element, such as a process or product. The questionnaire will utilise open-ended questions since it is not practical to adequately list every possible answer.

-closed question

Closed-ended questions are appropriate when the system analyst can comprehensively enumerate every potential response to the query and when each of the stated answers is mutually exclusive, meaning selecting one means selecting none of the others. closed-ended survey questions are applied to a sizable sample of respondents.

2. Data analysis

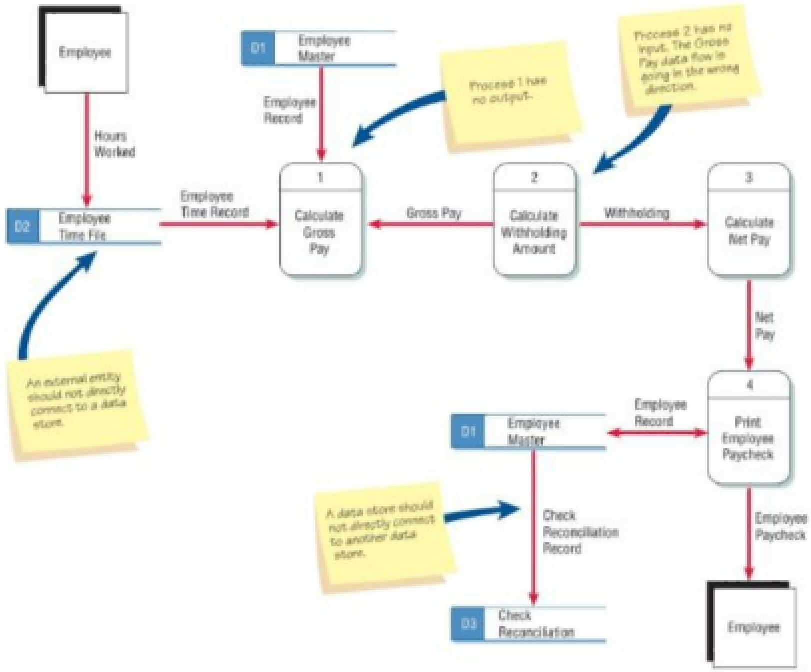
System analysts need to be able to visualise how data flows through an organisation, the processes or transformations it goes through, and the results it produces in order to try and comprehend the information needs of users. An organization's data operations can be represented graphically by the systems analyst using a structured analysis technique called data flow diagrams (DFDs).



Developing DFD

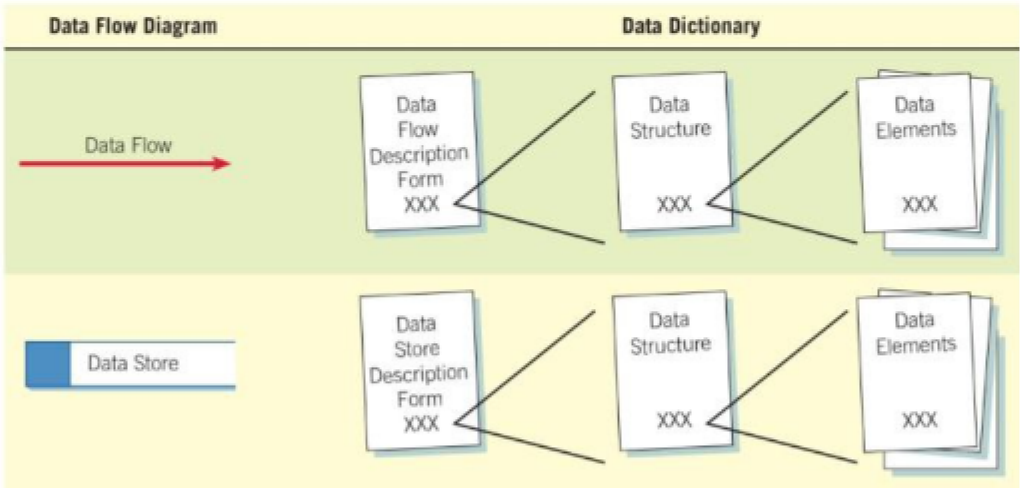
1. There must be minimum one process and must not have any stand alone objects or related to themselves
2. A process needs to generate at least one data flow out of it and receive at least one data flow into it.
3. There should be a minimum of one process linked to a data store.
4. There should be no connections between external entities. Even if they communicate on their own, the system constructed using DFDs does not include that communication.

Once the data flow diagrams are finished, the system analyst must work at giving the names of the data-oriented system's components unique meaning. Data-oriented system analysis is aided by a technique known as "data dictionary."



Data Dictionary

Data dictionary is a specialized application of the kinds of dictionaries used as references in everyday life (Kendall, 2010, p. 228). Systems analysts develop the data dictionary, a reference work on data about data, to help them with analysis and design. An automatic data dictionary is now a feature of many database management systems. These dictionaries might be straightforward or complex. Even with the automated data dictionary, the system analyst can still benefit from understanding the process of creating a data dictionary since it will help them visualise the system and how it functions.


Data Flow Diagram
Data Dictionary
Data Flow
Data Flow Description Form XXX
Data Structure XXX
Data Elements XXX
Data Store
Data Store Description Form XXX
Data Structure XXX
Data Elements XXX

4. UI design https://profagaskar.files.wordpress.com/2020/03/wiley_the_essential_guide_to_user_interf.pdf

User interface design is a subset of a field of study called human computer interaction (HCI). Human computer interaction is the study, planning, and design of how people and computers work together so that a user's needs are satisfied in the most effective way. The user interface is the component of the device and its software that users can see, hear, touch, converse or manipulate in any other way. Input and output are two main parts of the user interface. The way the computer communicates to the user the results of its operations and needs is called output. For users, a well-designed screen and interface are important. They act as both a doorway to the software's capabilities and their window to observe the system's capabilities. Although it is one of the few visible elements of the product its developers develop, it is the system to many consumers. Additionally, it provides a way via which multiple crucial tasks are presented. These responsibilities frequently directly affect an organization's revenue and consumer interactions. Methods for gaining an understanding of users (Gould, 1988) :

1. Visit user locations, particularly if they are unfamiliar to you, to gain an understanding of the user's work environment.
2. Talk with users about their problems, difficulties, wishes, and what works well now.
3. Establish direct contact; avoid relying on intermediaries.
4. Observe users working or performing a task to see what they do, their difficulties, and their problems.
5. Videotape users working or performing a task to illustrate and study problems and difficulties.
6. Learn about the work organization where the system may be installed.
7. Have users think aloud as they do something to uncover details that may not otherwise be solicited.
8. Try the job yourself. It may expose difficulties that are not known or expressed by users.
9. Prepare surveys and questionnaires to obtain a larger sample of user opinions.
10. Establish testable behavioral target goals to give management a measure for what progress has been made and what is still required.

5. Unified Modeling Language (UML) https://books.google.com.my/books?hl=en&lr=&id=1HpV_kVq7SUC&oi=fnd&pg=PR9&dq=unified+modeling+language+systems+analysis&ots=W_35cD4zoq&sig=-pWXa-p432EoUeMEWPGpDWbvRJI&redir_esc=y#v=onepage&q=unified%20modeling%20language%20systems%20analysis&f=false

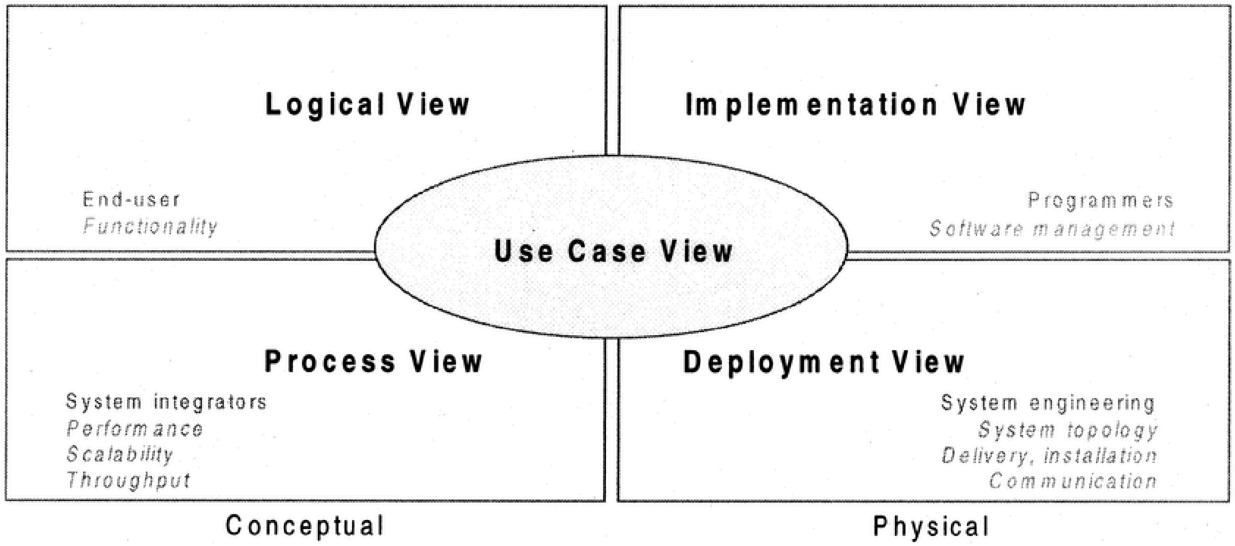
https://www2.ccs.neu.edu/research/demeter/course/f96/readings/uml9.pdf

Software engineers utilize Unified Modelling Language (UML), a standardized visual modelling language to design, visualize, define, develop, and document software-intensive systems. It offers a unified and consistent method for expressing the interactions, behaviors, and system architecture. UML is widely used in the software development process and is backes by a number of tools and techniques.

5.1 Modelling Architecture Views using UML https://www.google.com/url?sa=i&url=https%3A%2F%2Fwww.researchgate.net%2Ffigure%2FThe-4-1-architectural-views-present-an-architectural-design-addressing-the-needs-from_fig1_9072117&psig=AOvVaw1E73_-oxeRjKy8r0ZgjQIF&ust=1711986955069000&source=images&cd=vfe&opi=89978449&ved=0CBIQjRxqFwoTCMiAu9PunoUDFQAAAAAdAAAAABAH

Any real-world system is used by many users. Developers, testers, business professionals, analysts, and many more might be among the users. Therefore, several viewpoints need to be taken into consideration while creating the architecture of a system. Visualizing the system from several users' point of view is important. These perspectives are :

- Use Case View
- Design
- Implementation
- Process
- Deployment

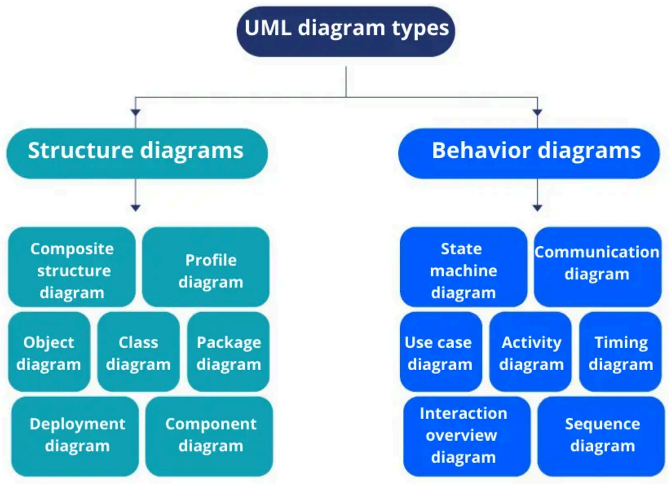


1. Use-case view : Describes what the system does, who uses it, and how it interacts with the external system. It's crucial for understanding system functionality and user interactions, deriving directly from the requirements.
2. Logical view : Shows how the system is organized internally, including classes, packages and interfaces. It illustrates relationships between these elements like dependencies and interfaces.
3. Implementation view : Explains how development files and artifacts are organized in the system's file structure. It covers both development and deployment artifacts, although it's not always mandatory for architectural understanding.
4. Process view : Illustrates how the system operates at runtime, including components, processes, threads, and interactions. It's vital for analyzing system behavior and performance, focusing on runtime quality attributes.
5. Deployment view: Details how the system is physically deployed onto hardware infrastructure. It's useful for understanding system-to-hardware mappings, although it's optional in some architectural frameworks.

5.2 14 Type of Diagrams in UML https://www.gleek.io/blog/uml-diagram-types

At UML's core are diagrams. These diagrams fall within the broad categories of behavioral and structural diagrams.

- Structural diagrams consist of static diagrams like class diagrams, objects diagrams, etc.
- Behavioral diagrams consist of dynamic diagrams like sequence diagrams, collaboration diagrams, etc.


UML diagram types
Structure diagrams
Behavior diagrams
Composite structure diagram
Profile diagram
Object diagram
Class diagram
Package diagram
Deployment diagram
Component diagram
State machine diagram
Communication diagram
Use case diagram
Activity diagram
Timing diagram
Interaction overview diagram
Sequence diagram

6. Comparative analysis

Traditional techniques like questionnaires, surveys, and interviews have long been used in the field of requirement analysis for obtaining user demands. These methods offer a moderate depth of insight with structured data gathering, making them adaptable to different project scales and types. In contrast, modern techniques like card sorting and laddering delve deeper into stakeholder perspectives and problem-solving strategies, offering a more nuanced understanding of user requirements. On the other hand, compared to traditional techniques, modern techniques might require a steeper learning curve and specialized tool assistance.

Traditional techniques of data analysis commonly involve statistical analysis, data mining, and qualitative analysis. Depending on the analytic approach, these strategies offer varying depth that may be adapted to the complexity of the data and the objectives of the study. However, while they need specialized tools and knowledge to execute, contemporary data analysis approaches like content analysis and theme analysis provide deeper insights into data patterns and relationships.

Traditional UI design techniques, such as User-Centered Design (UCD), place an intense focus on usability and user feedback to make sure interfaces successfully satisfy their needs. Because these techniques may be adapted to different user preferences and situations, they are frequently utilized in interface design. On the other hand, contemporary UI design approaches such as gamification, AR, and VUI (Voice User Interfaces) seek to improve user immersion and engagement, needing specific tools and design considerations.

Traditional techniques for UML usage require using standardized notation, such as class diagrams for system visualization and use case diagrams. Although these diagrams provide an organized visual representation of the needs and interactions of the system, they might not be suitable for complicated system modeling. Sequence diagrams and activity diagrams are examples of complex diagramming techniques used in modern UML use. These techniques offer extensive insights into system behavior, but they also need a higher degree of knowledge and tool support.

To summarize, traditional techniques give familiarity and flexibility, but modern techniques, however potentially challenging in terms of learning curve and tool needs, offer deeper insights and novel approaches in Requirement Analysis, Data Analysis, UI Design, and UML Usage.

Conclusion

In conclusion, the investigation of fundamental techniques in Data Analysis, UI Design, Requirement Analysis, and UML Usage has shed light on the constantly evolving field of system analysis and design. Traditional techniques have endured because they are dependable, flexible, and moderately in-depth when it comes to obtaining user requirements and doing data analysis. Modern techniques, on the other hand, have the potential to provide difficulties in terms of learning curve and tool needs, but they also bring innovation, deeper insights, and improved user engagement.

The importance of a balanced approach in system design and analysis is shown by this comparative analysis. Effective UI design, rapid data analysis, and an in-depth understanding of user requirements are made possible by the combination of traditional and modern techniques. Designers and developers may build systems that not only satisfy the needs of users but also innovate and improve the user experience by utilizing the advantages of each technique.

It will become important that practitioners and organizations continue to be adaptable and flexible in how they approach system analysis and design in the future. Using a variety of approaches, traditional and modern, empowers teams to address a wide range of problems, streamline workflows, and produce user-friendly solutions. In the end, successful, user-centric system development in today's changing technological context is made possible by a nuanced knowledge and strategic use of requirement analysis and design techniques.